Vanilla & modded Java clients

Default product path (`game-authority=folia`, `folia-jar-source=build`): YaP-Folia owns the public JE protocol after boot (not stock Fill). Legacy `game-authority=paper` uses Paper the same way. YaPcore's native multi-version bands apply when `game-authority=native`. See QUICK_START.md, CLIENTS_AND_PACKS.md, and CROSSPLAY.md.

YaPcore's native path speaks **real Minecraft Java protocol** with **built-in multi-version bands** (`ProtocolBand`). Each client gets its own codec path — no external translators.

Compatibility (default)

Setting	Default	Effect
`backwards-compatible`	`true`	Accept registered JE/BE protocol IDs; map unknowns to nearest band

Native bands cover **1.8 → 26.2**. Fabric/Quilt OK; Forge/NeoForge client noise is tolerated.

What can connect

Client	Expected
Vanilla	Server list ping + join (offline-mode)
Fabric / Quilt	Same as vanilla for client-side mods
NeoForge / Forge	Vanilla path with loader-compat

Requirements

```
online-mode=false
java-enabled=true
backwards-compatible=true
port=25566
```

1. Start (`./scripts/gui.sh` or `./scripts/start.sh`) — **you** start Link from the GUI/dashboard when using the proxy path
2. Connect: - **Through Link (product path):** `127.0.0.1:25565` - **Direct Via edge:** `127.0.0.1:25566`
3. MOTD / player counts should appear in the server list

Logs: STATUS ping vs join

Server-list refreshes use handshake **intent=1** (STATUS). The client closes after the ping — that is normal and is **not** logged as `JE JOIN FAILED` anymore. Actual joins use **intent=2** (LOGIN) and should reach PLAY.

Link join: `Frame length cannot be zero`

Vanilla rejects outer frames whose length VarInt is `0`. Link must send Set Compression **uncompressed**, then enable zlib on the **same** outbound handler that adds length prefixes (`McOutboundPacketEncoder`). After Link protocol changes, refresh **repo-root** `yap-link.jar` (GUI preference) via shadowJar + copy or `gradle publishReleasesFolder`.

Public domain / nginx: NETWORKING.md, CLOUDFLARE_AND_NGINX.md.

Configuration dumps (why joins used to fail)

Modern JE clients (1.20.2+) require a **vanilla-complete** Configuration phase: full synchronized registry ID lists + **all** Update Tags. Tags are never loaded from known packs — missing one (`infiniburn_*`, `sulfur_cube_archetype/*`, …) → client `Network Protocol Error` at Finish Configuration.

YaPcore ships official dumps under `src/main/resources/protocol/vanilla/<version>/`:

File	Source
`registryEntries.json`	Entry IDs from that release's server jar datapack
`networkTags.nbt`	Full Update Tags (IDs from `registries.json` + datapack)

26.2 packet dumps live under `src/main/resources/protocol/vanilla/` (checked in). When Mojang ships a newer protocol, regenerate dumps with `minecraft-data` / Mojang specs, commit `packets.json` + `index.json`, then commit under `src/main/resources/protocol/vanilla/`. Drop files in `protocol/vanilla/<ver>/` for additional versions.

Threading does not affect this path — join is protocol content, not YapEngine layout.

Pipeline

Native YaPcore JE: `McFrameCodec` → `JavaProtocolHandler` (client `ProtocolBand`) → `CrossplayHub`.

YaP Link JE: `McFrameCodec.Decoder` → (optional) `McCompressionCodec.Decoder` → session; outbound `McOutboundPacketEncoder` (zlib+length). See YAP_LINK.md.

Honest limits

- Chunk / entity sync is still minimal (teleport below world clears loading terrain).
- Online-mode Mojang auth not implemented (`online-mode=false`).
- Play packet IDs for very old eras are best-effort and improve over time.

See CLIENTS_AND_PACKS.md and CROSSPLAY.md.